

BUDGET-AWARE RUNTIME SAFETY MONITOR FUSION

Measurement Validity and Runtime Evaluation

Technical Report

Course: Trustworthy AI

Date: 15 July 2026

Repository commit: 68fe371

This report presents the final audited measurements and the resulting scientific interpretation of the project.

Abstract

This study evaluates a selective runtime safety-monitoring policy that uses inexpensive monitors to decide when an expensive Qwen3Guard call is necessary. The evaluation was rebuilt around an audited response-harmfulness target, pinned monitor implementations, serialized routing and fusion pipelines, common-FPR policy comparisons, nested source and attack-family holdouts, fixed-sequence Learn-then-Test risk control, and end-to-end GPU timing.

Selective acquisition reduced mean latency from 1654.37 ms to 1251.71 ms in a 128-example, batch-1 Tesla T4 benchmark, a reduction of 24.34%. The upper-tail measurements did not improve. More importantly, the prespecified 5% false-positive-rate condition did not hold reliably under the audited shift evaluations. In-fold risk-control certificates were valid for their untouched risk-control partitions but did not transfer to excluded sources or attack families. Two unique examples also changed Qwen classification scores when the pinned monitor was regenerated. The results support a measurement-validity interpretation rather than a routing-performance success claim.

1. Introduction

Runtime safety systems can combine fast, inexpensive filters with slower classifiers that provide richer safety judgments. A selective policy is useful only when the compute saved by skipping the expensive monitor does not come at an unacceptable safety cost. The question examined here was whether a learned routing stage could reduce Qwen3Guard acquisition while maintaining a false-positive-rate target of 0.05.

The comparison uses a full-information policy and a selective-acquisition policy. Full information always obtains the rule, compact, and Qwen scores before applying the fusion model. Selective acquisition first uses the cheap monitors and a serialized router. It allows, intercepts, or requests Qwen according to the frozen routing thresholds.

During validation, the central issue shifted from routing efficiency to measurement validity. The target concerns harmfulness of the model response, but prompt-only monitoring can react to harmful user content. The original final partition had also influenced development, and formal risk certificates did not transfer reliably to excluded sources and attack families. These findings determine how the final results should be interpreted.

2. System Design

The final v3 pipeline contains a deterministic rule filter, a compact text-moderation model, Qwen3Guard-Gen-4B, and calibrated logistic pipelines for fusion and routing. Feature order, preprocessing, fitted estimators, operating thresholds, and provenance were serialized so that evaluation and timing used the same locked policy.

Table 1. Monitor and policy components

Component	Implementation	Regeneration result
Rule filter	Deterministic rule_filter_v1	Exact score match
Compact monitor	KoalaAI/Text-Moderation, revision 838ade0e...	Maximum score difference 3.75e-7
Expensive monitor	Qwen/Qwen3Guard-Gen-4B, revision 6ec42827...	Official template and structured parsing; raw generations retained
Fusion and router	Serialized calibrated logistic pipelines	Reloaded bundle used for evaluation and timing

2.1 Policy execution

Both policies always compute the rule and compact-monitor scores. The full-information policy then calls Qwen for every example. The selective policy uses two router thresholds: low scores are allowed without Qwen, high scores are intercepted without Qwen, and scores in the uncertainty region proceed to Qwen and final fusion. This arrangement makes the expensive-call rate directly observable.

3. Data and Label Review

Table 2. Audited dataset and evaluation partitions

Item	Examples	Use
JailbreakBench	200	Adversarial jailbreak examples
XSTest	250	Benign over-refusal probes
WildGuardTest	1,709	Mixed safety-monitor examples
Unified dataset	2,159	Audited analysis set
Negative labels	1,756	Response judged not harmful
Positive labels	403	Response judged harmful
Train / selection	844 / 421	Model fitting and policy selection
Calibration / old final	422 / 422	Threshold diagnostics and descriptive evaluation
Held-out GCG	50	Additional shift diagnostic

3.1 Label audit

Two hundred rows were manually reviewed against the response-harmfulness definition. A second review examined 40 difficult or high-priority cases, and 19 labels were corrected. The audit improves alignment between the dataset and the stated target, but it is a single-author review rather than a multi-rater annotation study.

3.2 Monitor input modes

Qwen was evaluated with prompt-only, response-only, and prompt-response inputs. Prompt-response was used in the final pipeline because it retains the conversational context while directly including the response under evaluation. Prompt-only results were more strongly affected by harmful prompt content, which is inconsistent with treating them as a direct measure of response harmfulness. This is evidence of construct contamination, not proof of a single causal mechanism.

3.3 Status of the original final partition

The original final partition influenced development decisions and is therefore reported as descriptive evidence only. It is not treated as an untouched confirmatory test. The main out-of-sample diagnostics use nested leave-source-out and leave-family-out folds in which the outer examples were excluded from model fitting and threshold selection.

4. Experimental Method

Table 3. Evaluation procedures

Procedure	Implementation
Policy families	Six prespecified families selected on designated training and selection partitions, then frozen.
Common-FPR comparison	Full and selective policies compared at targets 0.01, 0.025, 0.05, and 0.10 with paired bootstrap intervals.
Nested holdouts	Leave-source-out folds for JailbreakBench, WildGuardTest, and XSTest; leave-family-out folds for GCG, PAIR, and random search.
Risk control	Within each outer fold: 40% model training, 20% candidate selection, and 40% untouched risk control.
GPU timing	Tesla T4, batch size 1, 128 deterministic examples, eight excluded warm-up rows, alternating policy order, CUDA synchronization.

4.1 Risk-control test

Candidate policies were ordered from conservative to aggressive. For each fold and method, a fixed-sequence exact binomial test evaluated the null hypothesis that FPR was at least 0.05. A policy was certified only when the one-sided 95% upper confidence bound on its untouched risk-control partition was no greater than 0.05. The certificate applies to that risk-control distribution; it does not automatically extend to the excluded outer group.

4.2 Stop/go criterion

The routing result was required to pass two gates: a cost advantage and a risk constraint. The project would be framed as a routing-performance result only if both conditions passed. The criterion was applied to the frozen v3 results without changing thresholds after evaluation.

5. Results

5.1 Stop/go result

Table 4. Prespecified stop/go decision

Gate	Outcome	Result
Cost advantage	Pass	Selective acquisition reduced expensive-monitor use and mean latency.
Risk constraint	Fail	The 0.05 FPR condition was not reliable under the audited shift evidence.
Overall decision	No-go	The routing-performance framing does not pass its prespecified rule.

The no-go decision does not indicate that the routing code failed. The selective mechanism produced a measurable efficiency gain, but that gain was not accompanied by sufficiently reliable false-positive control. The result is therefore better understood as evidence about measurement scope and transfer than as a successful routing-performance result.

5.2 Nested source and family holdouts

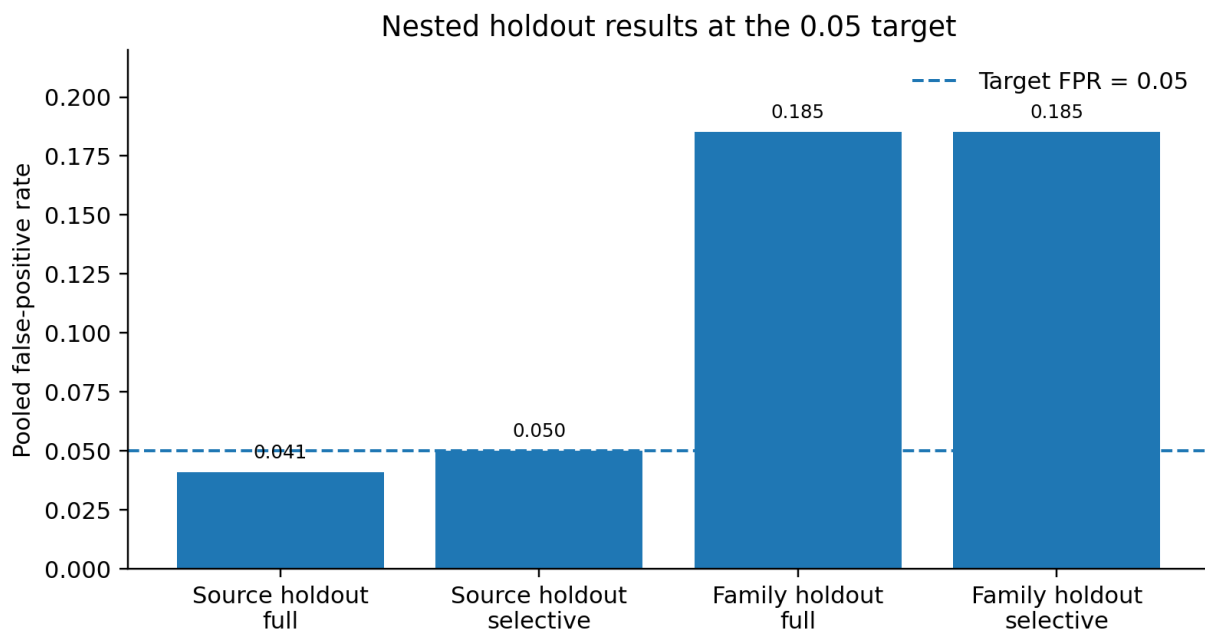


Figure 1. Pooled nested-holdout FPR at the 0.05 target. The dashed line marks the target.

Table 5. Pooled nested-holdout results at target FPR 0.05

Outer holdout	Policy	Recall	FPR	95% upper	Cost reduction
Source	Full information	0.82134	0.04100	0.04966	0
Source	Selective	0.83375	0.05011	0.05954	0.05494
Attack family	Full information	Not emphasized	0.18519	-	0
Attack family	Selective	Not emphasized	0.18519	-	0.00972

The pooled source-holdout result was close to the nominal target, but the selective-policy upper bound exceeded 0.05. The attack-family holdout produced a much larger FPR for both policies. These results show that the operating point learned on the available development data was not stable under all tested shifts.

5.3 Learn-then-Test certificates

All 12 fold-by-method sequences selected a policy whose one-sided upper bound was at most 0.05 on the untouched in-fold risk-control partition. The outer-group results nevertheless exceeded 0.05 in several cases. This distinction is important: the procedure controlled risk where it was tested, but the certificate did not establish transfer to a new source or attack family.

The fallback label used in the analysis is `selection_zero_intercept_fallback`. It means that the fallback intercepted no examples on the policy-selection partition. It does not mean that the same policy must intercept zero examples on later risk-control or outer partitions.

5.4 End-to-end timing

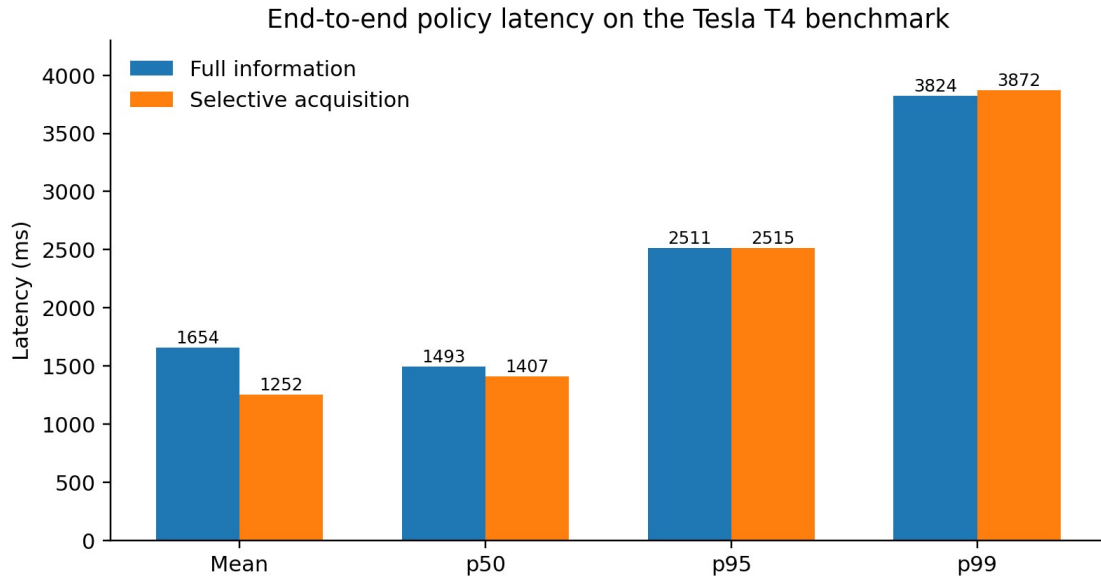


Figure 2. End-to-end latency for the full-information and selective policies.

Table 6. Final Tesla T4 timing summary

Policy or component	Mean ms	p50 ms	p95 ms	p99 ms
Rule filter	1.27	1.11	2.59	4.02
Compact monitor	45.66	41.63	70.66	76.61
Qwen prompt-response	1597.56	1441.37	2431.60	3744.09
Full information	1654.37	1493.17	2511.24	3824.36
Selective acquisition	1251.71	1407.09	2515.26	3872.32

Selective acquisition lowered mean latency by 402.66 ms, or 24.34%, and lowered the median by about 86.08 ms. It did not improve the upper tail: p95 increased by about 4.02 ms and p99 by about 47.96 ms. The timing result therefore supports an average-latency reduction in this benchmark, not a tail-latency improvement.

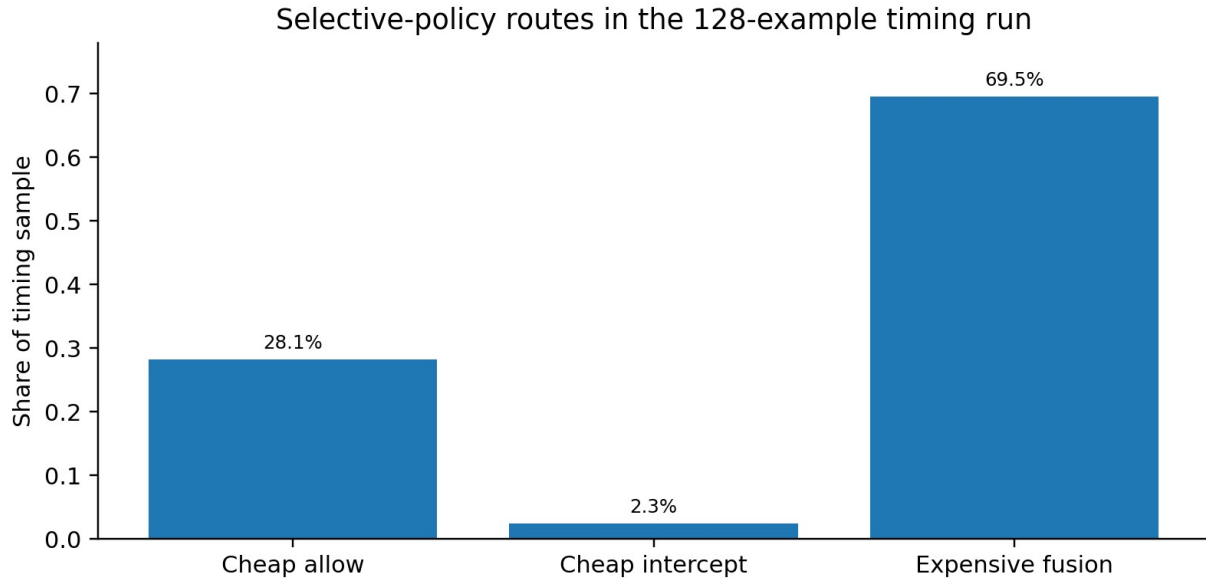


Figure 3. Route distribution for the selective policy in the timing benchmark.

Table 7. Selective-policy route distribution

Route	Examples	Rate	Mean latency
Cheap allow	36	28.13%	42.14 ms
Cheap intercept	3	2.34%	73.58 ms
Expensive fusion	89	69.53%	1780.69 ms

5.5 Regeneration reproducibility

Table 8. Score-regeneration checks from the final timing run

Monitor	Maximum difference	Additional result
Rule filter	0	Exact regeneration
Compact monitor	3.75e-7	Within numerical tolerance
Qwen3Guard	0.5	4 mismatch rows across 2 unique examples; 98.16% exact match

Every regenerated Qwen output was parsed successfully, but two unique examples changed discrete scores relative to the cached run. The raw generations and mismatch rows were retained. The timing values remain valid for the regenerated execution, while the score differences show that exact classification parity should not be assumed for this generative monitor.

6. Discussion

The project demonstrates why routing efficiency cannot be evaluated in isolation from the measurement process. The label definition, the text given to each monitor, the provenance of the evaluation split, the scope of a statistical certificate, and the reproducibility of a generative classifier all affected the final interpretation.

The latency reduction is practically meaningful within the measured setting, but it does not settle the safety question. The selective policy called Qwen on 69.53% of examples and achieved a lower mean runtime. Its risk behavior, however,

was not stable enough under source and attack-family shifts to support the required 5% FPR claim. The in-fold certificates and the outer-shift failures are compatible because they refer to different distributions.

The prompt-mode comparison and the Qwen regeneration mismatch also point to measurement concerns that would be hidden by reporting only a single policy metric. Prompt-only monitoring does not align cleanly with a response-level target, and parseable generation does not guarantee identical discrete classifications across runs. These observations make measurement validity the stronger contribution of the project.

6.1 Limitations

The benchmark includes three source datasets and three attack-family groupings and should not be treated as representative of all deployment traffic. The label review was conducted by one author rather than by multiple independent annotators. The original final split had already influenced development, so its results remain descriptive even after the nested analyses were added.

The timing sample contains 128 examples measured on one Tesla T4 at batch size 1. Results may differ on other hardware, batch sizes, or workloads. The source of the two Qwen score changes was not isolated, and the experiment does not establish how often similar changes would occur in other environments. Sentinel auditing was outside the completed scope.

7. Conclusion

The final implementation provides an audited response-harmfulness dataset, pinned monitor regenerations, serialized fusion and routing pipelines, common-FPR comparisons for the prespecified policies, nested source and family holdouts, fixed-sequence Learn-then-Test risk control, and end-to-end Tesla T4 timing.

Selective acquisition reduced mean latency by 24.34% in the measured benchmark, but the prespecified risk condition did not hold reliably under the tested shifts. The project therefore does not support a routing-performance success claim. Its stronger result is that target construction, monitor input choice, split provenance, certificate scope, distribution shift, and generative-monitor reproducibility can materially change the conclusion of a runtime safety evaluation.

On the current evidence, the appropriate next step is a measurement-validity paper that reports the efficiency result together with these limitations rather than presenting the selective route as a generally risk-controlled deployment policy.